Question 1

With respect to logistic regression, answer the following:

1. Logistic regression models the probability P(y = 1 | x) as a sigmoid function. Starting with the assumption that the log-odds (logit) is a linear function of x (i.e. $log\ (p/(1 - p)) = w^T x + b$), derive the equation for the sigmoid activation function $\sigma(z)$.
2. We do not arbitrarily choose "Log Loss" (Binary Cross Entropy). It is derived from the principle of Maximum Likelihood Estimation (MLE).
   - Assume the target y follows a Bernoulli distribution: $P(y|x) = \hat{y}^{y} (1 - \hat{y})^{(1-y)}$
   - Derive the Negative Log-Likelihood (NLL) loss function for a dataset of m examples.
3. In linear regression, we use mean squared error (MSE). Explain mathematically or visually why using MSE with a sigmoid activation function $L = 1/2\ (\hat{y} - y)^2$ is problematic for finding the global minimum.

Question 2

Consider the Kaggle Credit Card Fraud Detection dataset. This dataset is highly imbalanced.

- Train your logistic regression model on the dataset. Compute the accuracy, precision, recall, and F1 score. If your model achieves 99.8% accuracy, is it a good one?
- Standard logistic regression treats all errors equally. In fraud detection, a false negative (missing a fraud) is much worse than a false positive. Implement a **weighted log-loss.** Calculate the gradient for this new loss function and update your gradient descent rule. Set the weight for fraud much higher than the weight for normal (e.g., proportional to the class imbalance ratio).
- Plot the precision-recall curve using your predicted probabilities. Select a decision threshold that prioritizes **recall** over **precision**. What threshold value provides you with at least 80% recall?